

Program : Diploma in Polymer Technology	
Course Code : 4092	Course Title: Latex Technology
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To provide an idea about functions and dosages of various ingredients used in latex compounding.
- To be well versed on the principles, production processes, trouble shooting and quality control of dipped, foamed and extruded goods manufactured from latex.

Course Prerequisites:

Topic	Course code	Course name	Semester
Concentration of Natural Rubber Latex		Natural Rubber Production	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain different compounding ingredients and latex compounding	15	Understanding
CO2	Generalize the dipping and casting process for the manufacture of various products	15	Understanding
CO3	Paraphrase the production of latex foam and coir foam	14	Understanding
CO4	Explain the production of latex extruded products, adhesives and latex spreading	14	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3		3				
CO4	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain different compounding ingredients and latex compounding		
M1.01	State the major principles of latex compounding.	3	Understanding
M1.02	Distinguish between latex compounding and dry rubber compounding.	2	Understanding
M1.03	State the functions and dosages of the various ingredients used for latex compounding.	4	Understanding
M1.04	Explain the preparation of dispersions and emulsions of latex compounding ingredients.	4	Understanding
M1.05	Explain the major characteristics and applications of synthetic lattices used in latex products manufacturing	2	Understanding
Contents: Importance of natural rubber latex in product manufacturing. Compounding of latex - principles of latex compounding - difference between latex compounding and dry rubber compounding. Functions and dosages of vulcanizing agents - accelerators - activators - fillers - stabilizers, wetting agents - thickeners-gelling agents - heat sensitizing agents - emulsifiers - dispersing agents -antidegradants. Method of preparation of Zno - Sulphur - ZMBT - dispersion with typical formulations, working and parts of ball mill. Method of preparation of SP and oil emulsion. Characteristics and applications - synthetic lattices in latex product manufacture - SBR latex			
CO2	Generalize the dipping and casting process for the manufacture of various products		
M2.01	Explain the principle and procedure of latex dipping	4	Understanding
M2.02	Describe the manufacture of major dipped goods with formulation.	5	Understanding

M2.03	Describe major after treatments conducted to modify the properties of latex products.	3	Understanding
M2.04	Explain the principle and procedure of latex casting.	3	Understanding
	Series Test - I	1	
Contents: Dipping - principle and procedure of dipping process - different types of dipping - after treatment of latex deposits - Manufacture of dipped goods with formulation and flow chart - defects and remedies. latex casting - principle and procedure of casting - production of cast articles - mould preparation.			
CO3	Paraphrase the production of latex foam and coir foam		
M3.01	Explain the production of latex foam	5	Understanding
M3.02	Explain the production of coir foam	3	Understanding
M3.03	Describe the carpet backing process	3	Understanding
M3.04	Describe road surfacing with latex and latex cements.	3	Understanding
Contents: List latex foam products and their manufactures -method of foam production -Dunlop process and talalay process -different steps -principle of gelation and gelling agents - production of coir foam different steps -carpet backing -typical formulations. Latex cement -preparation-advantages formulation.			
CO4	Explain the production of latex extruded products, adhesives and latex spreading		
M4.01	Explain the manufacture of latex thread and tubings	5	Understanding
M4.02	Compare latex and solvent based adhesives	2	Understanding
M4.03	Describe the production of adhesives for paper, metal and glass	3	Understanding
M4.04	Illustrate the process of impregnation and spreading of latex	4	Understanding
	Series Test - II	1	
Contents: Latex extrusion - principle - formulation - equipment - process - products - equality control latex thread, Latex tubing. Latex adhesives - formulation - preparation - compare with solvent based adhesives. Different types of adhesives. latex spreading – formulation – impregnation - machinery - for latex spreading - different products			

Text / Reference:

T/R	Book Title/Author
T1	High Polymer Lattices, vol-I, D.C.Blackley (Publisher - SpringerNetherlands,1997)
R1	High Polymer Lattices, vol-II, - D.C.Blackley (Publisher - SpringerNetherlands,1997)
R2	High Polymer Lattices, Vol-III - D.C.Blackley (Publisher - SpringerNetherlands,1997)
R3	Latex in Industry - R.J.Noble (Publisher - Rubber Age, Palmerton, New York).
R4	Training Manual - Rubber Training Institute, RRII, Kottayam

Online Resources:

Sl.No	Website Link
1	https://www.jacionline.org
2	http://www.nocil.com
3	http://www.rubberscience.in .
4	https://www.researchgate.net
5	https://link.springer.com